

THE MORITA CONSTRUCTION AND MONOIDAL NERVES: A THESIS PROPOSAL

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1. INTRODUCTION

The overall goal of the thesis will be to investigate in detail the so-called Morita construction in the setting of monoidal category theory. A strict monoidal category is an ordinary category equipped with a strictly associative and unital product bifunctor. Monoidal categories provide the setting in which to define monoids (objects equipped with an associative and unital multiplication) and their modules (objects equipped with an action of a monoid on the left or right). A 2-category $\mathcal{C}$ consists of a collection of objects and ordinary categories $\mathcal{C}(a, b)$ for each pair of objects (a, b) whose objects and morphisms we call 1- and 2-morphisms respectively, along with a composition law which is functorial, associative, and unital. Given a strict monoidal category V , the Morita construction produces a 2-category in which the objects are the monoids in V , the 1-morphisms are bimodules, and the 2-morphisms are bimodule maps, with composition of 1-morphisms given by balanced tensor product.

The thesis will consist of a document (which I began writing under the guidance of Martina Rovelli in Fall 2024) which introduces the basic idea of the construction and the necessary background, establishes the basic results about the theory in the strict case, and works through some examples, proving everything along the way; as well as at least one joint paper with my faculty advisor and two of her PhD students on the general case of the theory, treated using 2-complicial sets as a model of $(\infty, 2)$ -categories. That is, in a general monoidal category, we do not have equality but rather specified natural isomorphisms between associated products and left/right units, subject to coherence laws,

and from such a monoidal category, we obtain not a strict 2-category, but an $(\infty, 2)$ -category with higher-dimensional morphisms which keep track of the extra isomorphisms. The document and paper are both about half-written right now, and to finish the thesis, I will fill in details and background as well as further results in the first document (as suggested by my faculty advisor), and assist in completing the joint paper.

This topic is exciting since monoidal categories are a super general and broadly applicable framework and the theory of monoids and modules (on which the Morita construction relies heavily) is the common abstraction of numerous ubiquitous pieces of modern mathematics. Exploring this construction in detail has been a great way to get comfortable with a robust and geometrically insightful model of (∞, n) -categories, known as simplicial sets. These are simplicial sets equipped with a marking to indicate which simplices are “equivalences”, satisfying several lifting properties.

2. REVIEW OF LITERATURE OR BIBLIOGRAPHY

2.1. Monoidal category background. The story told in this thesis begins with a review of the foundations of monoidal category theory. For a standard reference, see [ML98]. A *monoidal category* $\mathcal{C}^\otimes$ is a category $\mathcal{C}$ with a notion of “tensor product” given by a functor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ and a “unit object” $1 \in \mathcal{C}$, together with three natural isomorphisms:

- the *associator* $\alpha : (- \otimes -) \otimes - \Rightarrow - \otimes (- \otimes -)$
- the *left unitor* $\lambda : 1 \otimes - \Rightarrow \text{id}_{\mathcal{C}}$
- the *right unitor* $\rho : - \otimes 1 \Rightarrow \text{id}_{\mathcal{C}}$

satisfying the following coherence conditions: the triangle identity

$$\begin{array}{ccc} (x \otimes 1) \otimes y & \xrightarrow{\alpha_{x,1,y}} & x \otimes (1 \otimes y) \\ & \searrow \rho_x \otimes \text{id}_y \quad \swarrow \text{id}_x \otimes \lambda_y & \\ & x \otimes y & \end{array}$$

and the pentagon identity

$$\begin{array}{ccccc} & & (w \otimes x) \otimes (y \otimes z) & & \\ & \swarrow \alpha_{w,x,y \otimes z} & & \swarrow \alpha_{w \otimes x,y,z} & \\ w \otimes (x \otimes (y \otimes z)) & & & & ((w \otimes x) \otimes y) \otimes z \\ \uparrow \text{id}_w \otimes \alpha_{x,y,z} & & & & \downarrow \alpha_{w,x,y} \otimes \text{id}_z \\ w \otimes ((x \otimes y) \otimes z) & \xleftarrow{\alpha_{w,x \otimes y,z}} & & & (w \otimes (x \otimes y)) \otimes z \end{array}$$

(Already in this definition, we see some hints of the combinatorial complexity that comes along with higher categorical structures. Monoidal categories, being bicategories with one object, require us to keep track of the associator 2-morphism, which comes with its own coherence law in the form of the associahedron K_4 , in addition to the unitors and their coherence law. It’s not that bad *yet* at this level, but it’s an omen of what’s to come.)

Monoidal categories are ubiquitous throughout modern mathematics for a variety of reasons. For one, monoidal categories provide the ideal setting in which one can discuss “processes in series *and in parallel*”. This perspective is given an inspiring exposition

in [Bra18]. In any category $\mathcal{C}$, one might interpret the objects x as states of a system, the morphisms $f : x \rightarrow y$ as processes transforming one state x into another y , and composition $g \circ f : x \rightarrow z$ of morphisms $f : x \rightarrow y$ and $g : y \rightarrow z$ as “doing g after f in series”. A monoidal product structure $\otimes$ on $\mathcal{C}$ provides exactly the notion of parallelism: for $f : w \rightarrow x$ and $g : y \rightarrow z$, the morphism $f \otimes g : w \otimes y \rightarrow x \otimes z$ is interpreted as “doing the processes f and g simultaneously, in parallel”. The coherence data ensure this makes sense: the associator, for instance, ensures that (up to natural isomorphism) a bunch of processes in parallel have only the data of a list, not the order in which they were added to the list, and the unit ensures there’s a “do-nothing process” which doesn’t change anything when you do it in parallel with others.

But for us, monoidal categories hold significance for the following reason: they are the universal setting in which to define monoids and modules over them. Classically, a monoid would usually be defined as a set X , together with a binary operation $\cdot$ on it and an element $1 \in X$ such that for all $x, y, z \in X$, we have $1 \cdot x = x = x \cdot 1$ and $(x \cdot y) \cdot z = x \cdot (y \cdot z)$. It’s meant to capture as bare-bones a notion of multiplication as one can have while still remaining useful. We could rephrase this definition in categorical language as follows: a monoid is an object $X \in \mathbf{Set}$ together with a morphism $\mu : X \times X \rightarrow X$ (this is $\cdot$) and a morphism $\eta : \{*\} \rightarrow X$ (this is 1), such that the following diagrams commute:

$$\begin{array}{ccccc} \{*\} \times X & \xrightarrow{\eta \times \text{id}_X} & X \times X & \xleftarrow{\text{id}_X \times \eta} & X \times \{*\} \\ & \searrow \lambda_X & \downarrow \mu & \swarrow \rho_X & \\ & & X & & \end{array}$$

and

$$\begin{array}{ccc} (X \times X) \times X & \xrightarrow{\alpha_{X,X,X}} & X \times (X \times X) \\ \mu \times \text{id}_X \downarrow & & \downarrow \text{id}_X \times \mu \\ X \times X & \xrightarrow{\mu} X \xleftarrow{\mu} & X \times X \end{array}$$

But what if we want to talk about multiplication in contexts with more structure than plain old sets? Fortunately, now that we’ve phrased our definition in categorical language, with no reference to the specifics of $\mathbf{Set}$, we can easily generalize this notion to any monoidal category $\mathcal{C}^\otimes$: a monoid in $\mathcal{C}^\otimes$ consists of an object $A \in \mathcal{C}$, a “multiplication” morphism $\mu : A \otimes A \rightarrow A$, and a “unit” morphism $\eta : 1 \rightarrow A$, such that the following diagram commutes:

$$\begin{array}{ccccc} 1 \otimes A & \xrightarrow{\eta \otimes \text{id}_A} & A \otimes A & \xleftarrow{\text{id}_A \otimes \eta} & A \otimes 1 \\ & \searrow \lambda_A & \downarrow \mu & \swarrow \rho_A & \\ & & A & & \end{array}$$

i.e. multiplication is unital, and this diagram also commutes:

$$\begin{array}{ccc}
 (A \otimes A) \otimes A & \xrightarrow{\alpha_{A,A,A}} & A \otimes (A \otimes A) \\
 \mu \otimes \text{id}_A \downarrow & & \downarrow \text{id}_A \otimes \mu \\
 A \otimes A & \xrightarrow{\mu} A \xleftarrow{\mu} & A \otimes A
 \end{array}$$

i.e. multiplication is associative. There is an easy-to-define notion of monoid homomorphism, or map of monoids.

Taking $\mathcal{C}^\otimes = \mathcal{Set}^\times$ recovers our above classical definition. Taking it to be, for example, the category of Abelian groups with tensor product over $\mathbb{Z}$, we recover the usual notion of rings, arguably the central object of study in algebra. Taking instead $\mathcal{Set}$ with coproduct $\amalg$ instead of product $\times$, it turns out we recover exactly sets themselves (and in fact, this is true for any cocomplete category).

Monoids encode the abstract idea of (associative, unital) multiplication. But these abstract structures are not of much use without the ability to act on concrete objects of interest. A *(left) module* over a monoid A in a monoidal category $\mathcal{C}^\otimes$ is an object $M \in \mathcal{C}$ together with a “left action” morphism $\ell : A \otimes M \rightarrow M$ which satisfies similar kinds of “unitality” and “associativity” conditions to those in the definition of monoids (for brevity, we won’t include every diagram here). There is a similar definition of right A -module, with a morphism $r : M \otimes A \rightarrow M$. There are easy-to-define notions of (left/right) A -module homomorphisms, or maps of modules.

These definitions encode the notion of “a thing on which a monoid acts multiplicatively”. For example, in $\mathcal{Ab}^{\otimes \mathbb{Z}}$, we recover the traditional notion of module over a ring, and in $\mathcal{Set}^\amalg$, we recover simply functions $X \rightarrow Y$.

We will be most interested in objects M with both a left A -module and right B -module structure for some monoids A and B , such that the two actions commute in the obvious way. These are known as (A, B) -bimodules, and are often denoted ${}_A M_B$. Bimodules admit a construction which is the most general incarnation of the notion of tensor product, called the “balanced tensor product”. Given a bimodule ${}_A M_B$ and another ${}_B N_C$, we define their *balanced tensor product over B* as the coequalizer of the following parallel pair:

$$(M \otimes B) \otimes N \begin{array}{c} \xrightarrow{r_M} \\ \xrightarrow{\ell_N \circ \alpha_{M,B,N}} \end{array} M \otimes N$$

This coequalizer, denoted $M \otimes_B N$, is what we obtain when we demand that there is no difference between B acting on M from the right, and B acting on N from the left. One can verify that the bimodule structures on M and N yield an (A, C) bimodule structure on $M \otimes_B N$. This brings us to the first level of the construction which is the main focus of this thesis.

2.2. The strict Morita category. In the early 1950s, Kiiti Morita proved a series of structure theorems regarding rings, modules, and their endomorphism rings (unpublished). In particular, he introduced and studied a new form of equivalence between rings, known as “Morita equivalence”. Two rings R, S are Morita equivalent if they have (additively) equivalent categories of modules ${}_R \mathcal{Mod}$ and ${}_S \mathcal{Mod}$. His work was popularized in a series of lectures in 1962 by Hyman Bass [Bas62], whereafter the subject became known as “Morita theory”. It is believed that Morita’s results should hold more generally for

monoids in any reasonably well-behaved monoidal category. One consequence of Morita’s work is that Morita equivalence is, in fact, simply isomorphism in a certain category known as the Morita category. This category and its refinements will form our main objects of study in this thesis, and one of our main questions to answer will be to determine when exactly a monoidal category is “well-behaved” enough for this construction to work.

For the rest of this section, let $\mathcal{C}^\otimes$ be a *strict* monoidal category, that is, one in which the associator and unitors are all identities. We say $\mathcal{C}^\otimes$ *admits a calculus of balanced tensor products* if $\mathcal{C}$ has all balanced tensor product coequalizers and $\otimes$ preserves them. We verify: so long as $\mathcal{C}^\otimes$ admits a calculus of balanced tensor products, there exists a 1-category, the *Morita category* $\mathfrak{M}(\mathcal{C}^\otimes)$ of $\mathcal{C}^\otimes$, which has as objects the monoids A in $\mathcal{C}^\otimes$, as morphisms $A \rightarrow B$ the bimodules ${}_A M_B$ in $\mathcal{C}^\otimes$, with composition of ${}_A M_B$ and ${}_B N_C$ given by the balanced tensor product ${}_A M \otimes_B N_C$. It is in this setting that two monoids are isomorphic precisely when they are Morita equivalent, and it is the above hypothesis that we argue is the loosest set of requirements on $\mathcal{C}^\otimes$ such that the Morita construction works. But where are the maps of bimodules?

For this, we will need to introduce another dimension. A *2-category* is a category enriched over *Cat* itself. Unpacking this, since the hom-sets are really hom-categories and composition is functorial on these, we really have two levels of morphisms with two (compatible) composition laws. We call the objects of the hom-categories “1-morphisms” and the morphisms of the hom-categories “2-morphisms”. We can compose 1-morphisms as before, but we can also do so with 2-morphisms, and we call this “horizontal composition” in contrast with the composition law on 2-morphisms in each hom-category, called “vertical composition”. Compositions of 2-morphisms are often represented using “globular pasting diagrams” such as the following:

$$\begin{array}{ccccc}
 & & f & & \\
 & \nearrow & \Downarrow \varphi & \searrow & \\
 A & \xrightarrow{\quad} & B & \xrightarrow{\quad h \quad} & C \\
 & \searrow & \Downarrow \psi & \nearrow & \\
 & & k & &
 \end{array}$$

The above picture makes the geometric point clear: objects are 0-dimensional, 1-morphisms are 1-dimensional, and we have added a second dimension by incorporating 2-morphisms.

Having said this, we can now upgrade our first construction to the second dimension: under the same hypotheses, there is a Morita 2-category $\mathfrak{M}(\mathcal{C}^\otimes)$, in which our 2-morphisms are bimodule maps, with vertical composition being the composition in $\mathcal{C}$.

2.3. The Morita $(\infty, 2)$ -category. The constructions of the previous section were nice and simple, but they relied on the assumption that $\mathcal{C}^\otimes$ was a *strict* monoidal category, which is not of terribly great use to us. Most naturally-occurring monoidal categories are not strict - even in $\mathbf{Set}^\times$, we do not have $(X \times Y) \times Z = X \times (Y \times Z)$. So we need to keep track of the associator and unitors. But it turns out that we cannot do this in an ordinary 2-category without losing information in a sloppy way (for example, by taking isomorphism classes of bimodules as our 1-morphisms). This problem demands the use of bona-fide *weak higher categories*, in our case, a so-called $(\infty, 2)$ -category.

Philosophically, for any $0 \leq n \leq \infty$, an (∞, n) -category is a structure reminiscent of a category, with compatible compositions of some sort in each dimension $k \geq 0$, with

coherence laws for these compositions holding only up to higher equivalence, and in which all morphisms in dimensions $> n$ are invertible in some sense. There exist quite a few competing models of these, all of which are in a certain (model-theoretic) sense equivalent, and each of which has various advantages and disadvantages. For our purposes, we will employ the so-called “complicial sets” model of (∞, n) -categories, introduced by Dominic Verity [Ver08]. Substantial research on them has been carried out by Emily Riehl, who also wrote the standard reference for gaining familiarity with them [Rie16]. My faculty mentor for this project, Martina Rovelli, along with her collaborator Viktoriya Ozornova, also complemented Verity’s vision by verifying the existence of a model structure on marked simplicial sets in which n -complicial sets are precisely the fibrant objects, in their paper [OR20].

Complicial sets are certain kinds of marked simplicial sets. A simplicial set is given by a functor $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$, where Δ is the category of finite ordinals with monotone maps. But really, one thinks of them as “triangulated spaces” of sorts, in which $X(k)$ consists of the (possibly degenerate) k -dimensional triangles (simplices) in the space. A marked simplicial set is one equipped with a choice $X(k)' \subseteq X(k)$ of k -simplices which we say are “marked”, for each $k \geq 1$. An n -complicial set is one of these possessing several lifting properties which essentially guarantee that k -simplices can be “composed” in a coherent way, and such that all simplices in dimensions $> n$ are marked. Here, marking of simplices captures the notion of “being an equivalence”.

In the latter section of this thesis, joint with Arghan Dutta, Stefano Luneia, and Martina Rovelli, we prove the following: for any monoidal category $\mathcal{C}^{\otimes}$ admitting a calculus of balanced tensor products, there exists a 2-complicial set $\mathfrak{M}(\mathcal{C}^{\otimes})$ in which the 0-morphisms are monoids in $\mathcal{C}^{\otimes}$, the 1-morphisms are bimodules, and the 2-morphisms are bimodule maps. We give the marking and the 3-morphisms in a natural way, and carry out the long-winded verification of the complicial set axioms. In a further paper, we will take any *braided* monoidal category $(\mathcal{C}^{\otimes}, \beta)$ and construct from it a “monoidal $(\infty, 2)$ -category” in the form of a 3-complicial set with one isomorphism class of objects.

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3. METHOD/APPROACH/PROCESS

Once a specific question is identified, (theoretical) mathematical research is generally carried out by trying to understand the concepts and constructions relevant to the question via worked examples and computations, conjecturing and proving lemmas and properties about them, and finally finding (through reasoning and trial/error) and carrying out a rigorous proof of the desired result. This is what we have done for this project. Typically we would meet around once per week to discuss progress and difficulties in the investigation, and plan our next directions. There is no data to be collected or special training required; most of the work is done in front of a chalkboard, or reading through background/reference material.

4. EVALUATION

Throughout earlier work on the project, incremental progress was evaluated and discussed in weekly meetings with Martina. The goals we've agreed upon for the final thesis are to finish the first of the two joint papers we're working on as a group (with the two PhD students), and to fill in details, background, and exposition in the document I produced while working with Martina last fall. We will meet regularly to ensure progress is being made at a reasonable pace.

5. COMMUNICATION

Given that Martina will spend the next academic year at the University of Ottawa, we will stay in touch over email/Discord/Zoom meetings and communicate regularly (perhaps weekly) about the project. This is roughly what we have been doing for a while now, and we have not had any issues with this setup. I plan to work for perhaps 20 hours/week on this project, outside of meetings.

6. TIMELINE

As suggested above, we will review results roughly weekly. Most of the material is already in a somewhat advanced draft state, and we will continue to manage the timing of the project as work progresses. According to PATHS, the final written component must be submitted by Dec 15 2025, so we'll try to have it done by perhaps Dec 8. We'll schedule the final presentation of the work later in the semester; it will most likely be somewhat informal.

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